

SHIV NADAR UNIVERSITY CHENNAI

School of Engineering

CS3807 – Deep Learning Laboratory

Experiment – 4

Comparative Study of Deep CNN Architectures Using Transfer learning

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Experiment 3

Comparative Study of Deep CNN Architectures using transfer learning

1. Objective

The objectives of this experiment are

- Study the evolution of deep CNN architectures.
- Compare LeNet-5, AlexNet, VGG16, GoogleNet and ResNet.
- Understand transfer learning.
- Fine tune pretrained CNN models.
- Compare classification performance of different architectures.

2. Learning Outcomes

After completing this experiment students will be able to

1. Explain modern CNN architectures.
2. Differentiate LeNet, AlexNet, VGG16, GoogleNet and ResNet.
3. Implement transfer learning.
4. Fine tune pretrained CNN models.
5. Compare CNN architectures using performance metrics.

3. Introduction

Deep Convolutional Neural Networks have become the dominant models for computer vision applications because they automatically learn hierarchical features from images. The evolution of CNNs has focused on improving classification accuracy while reducing computational complexity and training difficulties.

The progression of CNN architectures began with LeNet-5 and has evolved through AlexNet, VGGNet, GoogleNet, and ResNet. Each architecture introduced important innovations that significantly improved image classification performance.

4. Evolution of CNN Architectures

Model	Year	Major Contribution
LeNet-5	1998	First practical convolutional neural network
AlexNet	2012	ReLU activation, Dropout and GPU training
VGG16	2014	Uniform 3×3 convolution filters
GoogleNet	2014	Inception modules for multi-scale feature extraction
ResNet	2015	Residual learning using skip connections

5. LeNet-5

LeNet-5 was proposed by Yann LeCun in 1998 for handwritten digit recognition. It consists of two convolution layers, two average pooling layers and fully connected layers. It demonstrated that convolutional networks could automatically learn image features without handcrafted feature extraction.

Advantages

- Small number of parameters
- Fast training
- Suitable for OCR

6. AlexNet

AlexNet won the ImageNet Challenge in 2012 by reducing the classification error significantly. It introduced ReLU activation, Dropout regularization and GPU training.

Major innovations

- ReLU activation
- Dropout
- GPU implementation
- Data augmentation

7. VGG16

VGG16 demonstrated that using multiple small 3×3 convolution filters could outperform larger filters while increasing network depth.

Advantages

- Excellent feature extraction

- Simple architecture
- High classification accuracy

8. Comparison of Architectures

Architecture	Depth	Parameters	Main Contribution
LeNet-5	7	60K	First CNN
AlexNet	8	61M	ReLU + Dropout
VGG16	16	138M	Deep 3×3 filters
GoogLeNet	22	6.8M	Inception Modules
ResNet50	50	25.6M	Residual Learning

9. GoogLeNet (Inception Network)

GoogLeNet, proposed by Szegedy *et al.* in 2014, introduced the **Inception Module**, which performs multiple convolution operations in parallel using different filter sizes. Instead of using only a single kernel size, GoogLeNet simultaneously applies 1×1 , 3×3 , 5×5 convolutions and max pooling. The outputs are concatenated to obtain multi-scale feature representations.

The use of 1×1 convolutions before larger kernels reduces the number of trainable parameters and computational complexity.

Advantages

- Multi-scale feature extraction.
- Fewer parameters than VGG16.
- High computational efficiency.
- Better classification performance.

10. ResNet

Increasing CNN depth often causes the **vanishing gradient problem**, making optimization difficult. ResNet solves this issue using **skip (residual) connections**, allowing gradients to flow directly through the network.

Instead of learning

$$H(x)$$

ResNet learns

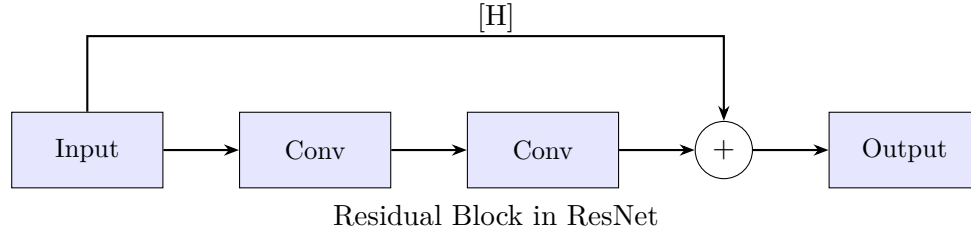
$$F(x) = H(x) - x$$

thus,

$$Output = F(x) + x$$

where

- $F(x)$ = Residual Mapping
- x = Identity Mapping



Advantages

- Eliminates vanishing gradients.
- Enables very deep neural networks.
- Faster convergence.
- High ImageNet accuracy.

11. Dilated Convolution

Dilated convolution (also called atrous convolution) increases the receptive field without increasing the number of parameters.

The dilation rate determines the spacing between kernel elements.

For a standard convolution,

$$D = 1$$

For dilated convolution,

$$D > 1$$

Example

$$\begin{bmatrix} 1 & 0 & 2 \\ 0 & 3 & 0 \\ 4 & 0 & 5 \end{bmatrix}$$

where zeros indicate inserted gaps.

Applications

- Semantic segmentation
- Medical image analysis
- Satellite imagery
- Object localization

12. Transpose Convolution

Transpose convolution increases the spatial dimensions of feature maps. Unlike pooling, which reduces image size, transpose convolution performs learnable upsampling.

Applications

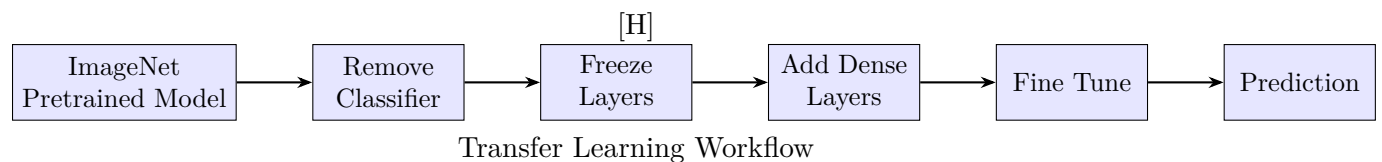
- Image Super Resolution
- Autoencoders
- GANs
- Image Generation
- Semantic Segmentation

13. Transfer Learning

Transfer Learning utilizes a pretrained CNN model that has already learned general image features from a large dataset such as ImageNet.

Instead of training from scratch,

1. Load a pretrained model.
2. Remove the original classifier.
3. Freeze convolution layers.
4. Add new Dense layers.
5. Train the classifier.
6. Fine tune selected convolution layers.



Advantages

- Faster convergence.
- Higher accuracy on small datasets.
- Reduced training time.
- Lower computational cost.

14. Dataset

Dataset: CIFAR-10

- Training Images : 50,000
- Testing Images : 10,000
- Number of Classes : 10
- Image Size : $32 \times 32 \times 3$
- Classes: Airplane, Automobile, Bird, Cat, Deer, Dog, Frog, Horse, Ship and Truck.

15. Experimental Procedure

Task 1: Dataset Preparation

1. Load the CIFAR-10 dataset using TensorFlow/Keras.
2. Normalize the pixel values to the range $[0,1]$.
3. Display ten sample images.
4. Print the dimensions of the training and testing datasets.

Task 2: Transfer Learning

Load any one of the following pretrained CNN models.

- VGG16
- ResNet50
- MobileNetV2
- EfficientNetB0 (Optional)

Perform the following steps.

1. Load pretrained ImageNet weights.
2. Remove the original classification layer.
3. Freeze the convolutional base.
4. Add a Global Average Pooling layer.
5. Add a Dense layer with ReLU activation.
6. Add the output layer with Softmax activation.

Task 3: Model Training

Train the model using

Optimizer	Adam
Learning Rate	0.001
Batch Size	32
Epochs	10–20
Loss Function	Categorical Cross Entropy
Evaluation Metric	Accuracy

Task 4: Fine Tuning

1. Unfreeze the last convolution block.
2. Train for another 5–10 epochs.
3. Compare the classification accuracy before and after fine tuning.

Task 5: Model Evaluation

Evaluate the trained model using

- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix
- Classification Report

16. Hyperparameter Study

Compare the performance using different settings.

Hyperparameter	Values
Learning Rate	0.001, 0.0001
Batch Size	16, 32, 64
Epochs	10, 20
Optimizer	Adam, SGD
Dense Units	128, 256
Frozen Layers	All, Partial

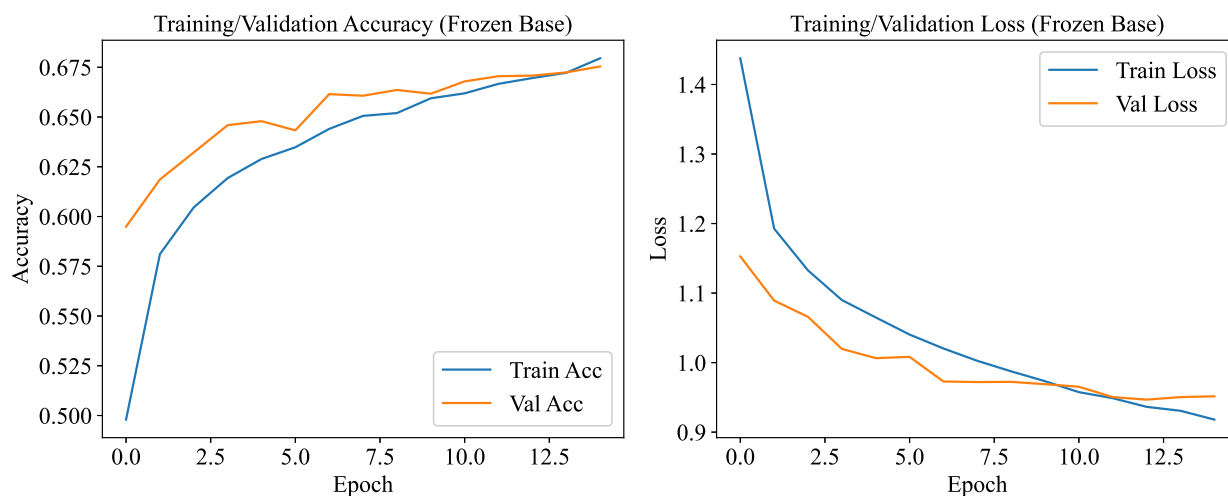
17. Mandatory Plots

17.1 Sample CIFAR-10 images

Sample Training Images

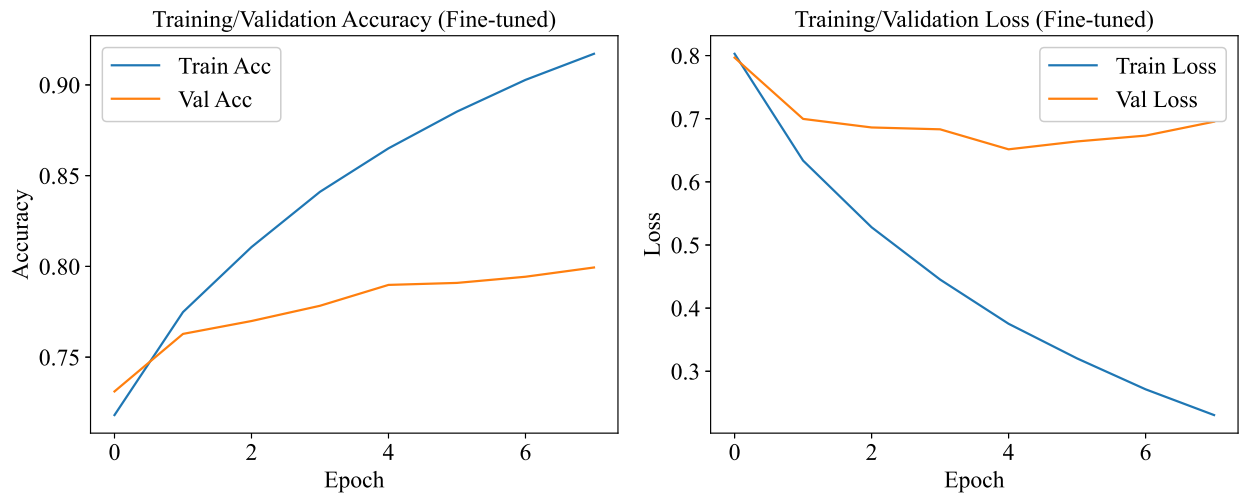


17.2 Training Curves (Frozen)



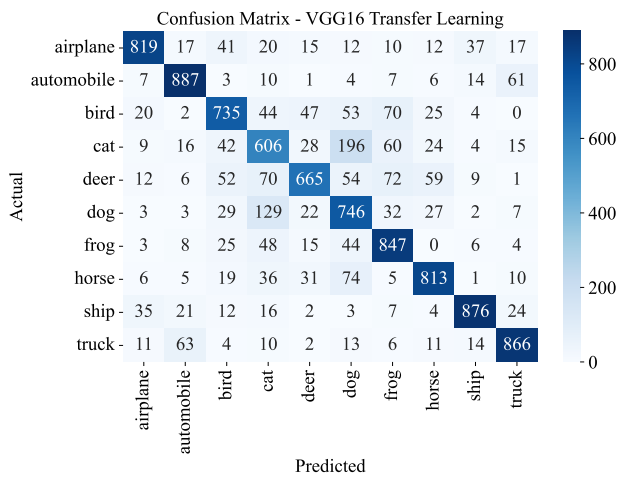
Accuracy rises steadily and loss falls smoothly without divergence . However frozen VGG16 features are underfitting since only the small dense head is trainable.

17.3 Training Curves (Fine Tuned)



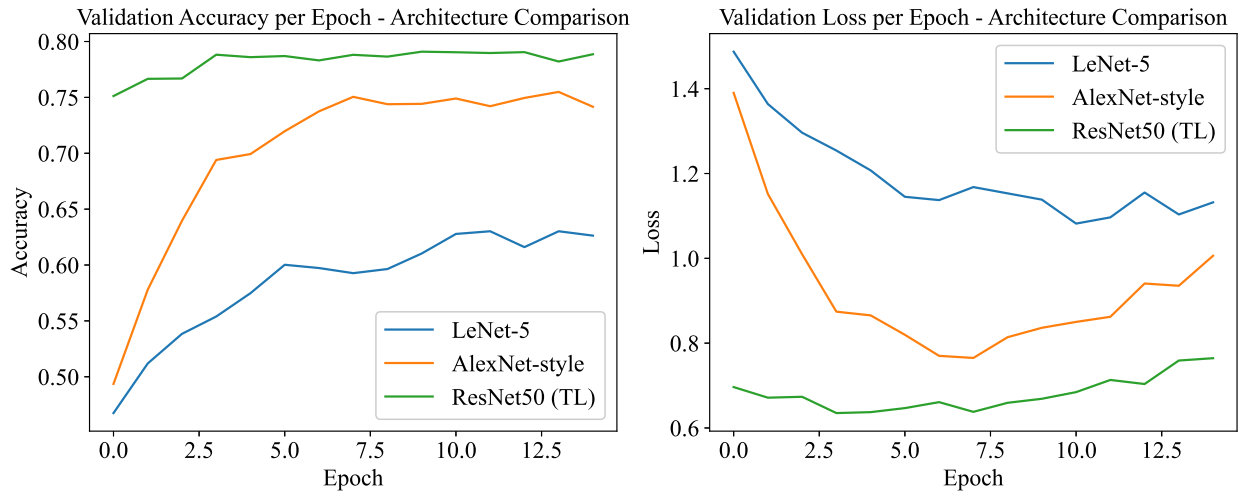
Accuracy and Loss is way better when compared to the frozen version but the model seemed to highly overfit as the training accuracy is over 0.9

17.4 Confusion Matrix



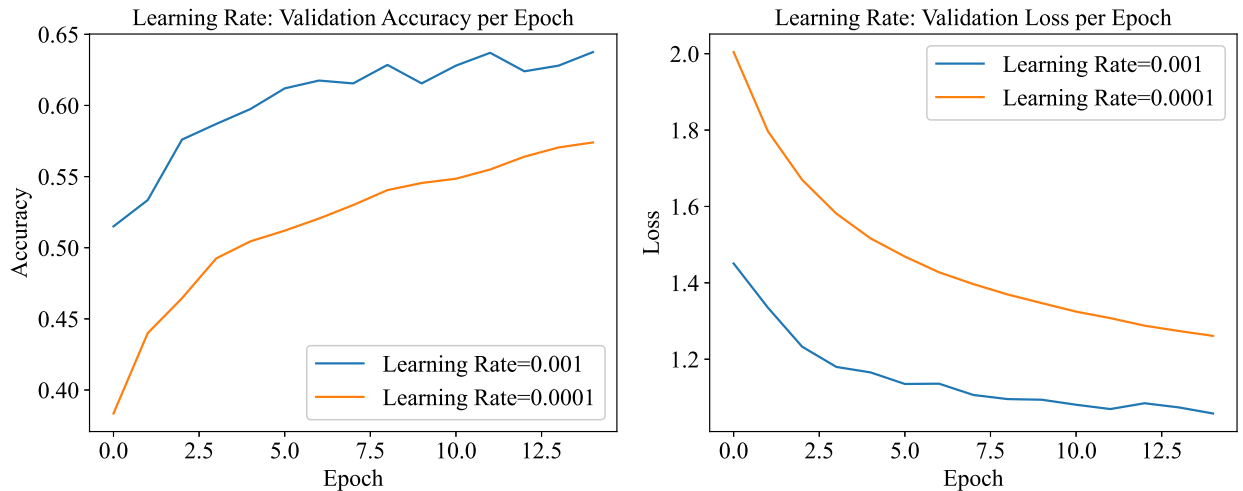
Overall the true positives are high in count suggesting good accuracy but still the visually similar objects like dogs, cats and confusion among vehicles still exist

17.5 Comparison LeNet AlexNet ResNet



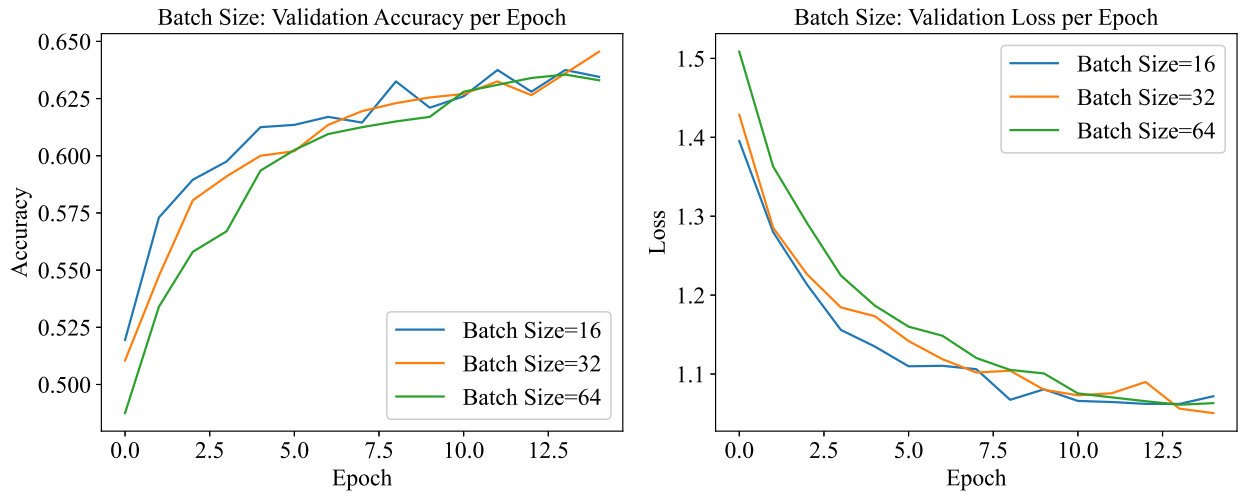
Resnet performed better than AlexNet better than Lenet in accuracy . Which suggests transfer learning (Resnet) performs better than a shallow (Lenet) and a deeper (Alexnet) architectures in this small dataset

17.6 Comparison Learning Rate (0.001, 0.0001)



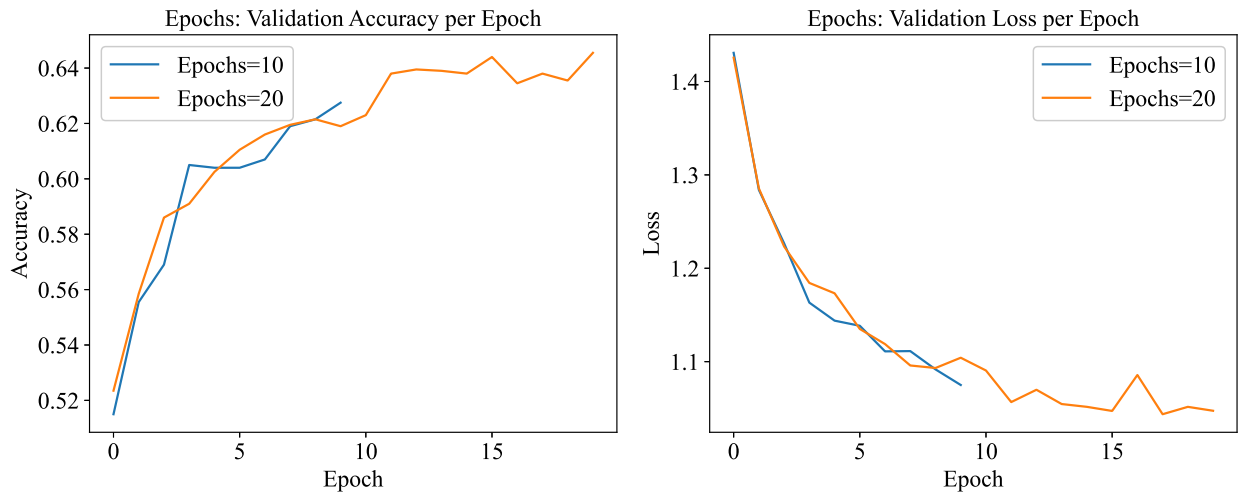
Higher learning 0.001 converges faster and reaches higher val accuracy than the very low learning rate 0.0001

17.7 Comparison Batch Size (16,32,64)



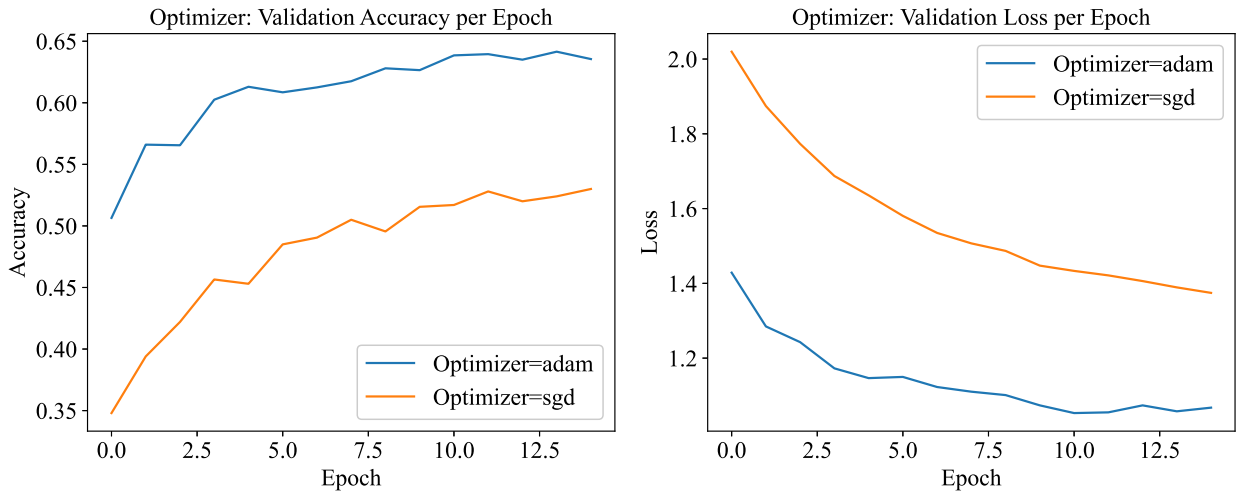
Curves for all the above batch sizes are highly similar

17.8 Comparison Epochs (10,20)



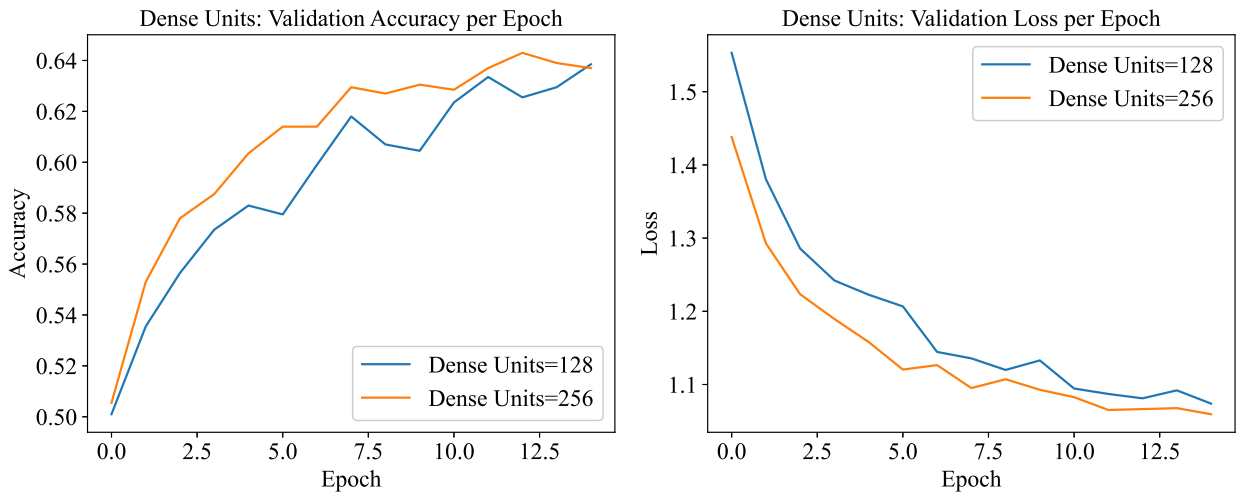
As the epoch count continues after 10 both the accuracy has risen and the loss has been reduced but flattened towards the end of it

17.9 Comparison Optimizer (Adam , SGD)



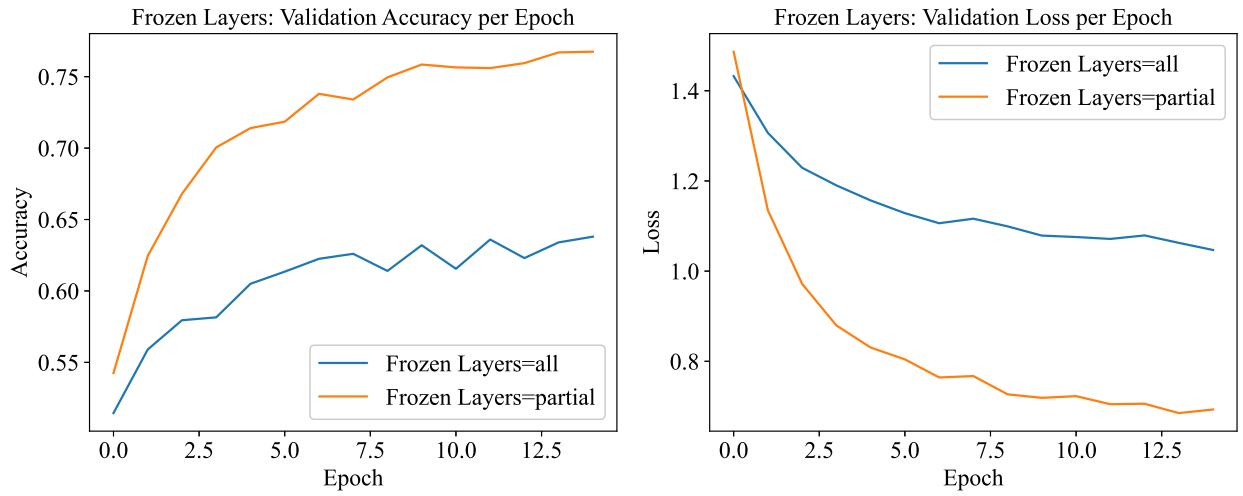
Adam converges faster and reaches higher accuracy than SGD within the same epoch budget

17.10 Comparison Dense Units (128 , 256)



256 units performs slightly better than 128 implying a larger dense head has slightly more capacity to combine the frozen VGG16 features, though the gap is likely small since the bottleneck is the frozen backbone and not head size.

17.11 Comparison Frozen Layers (partial , all)



The partial frozen layers version performs way better than the all frozen layers due to the finetunign(partial) happening at the cost of the additional training time

18. Results

18.1 Performance Metrics

Metric	Value
Training Accuracy	91.72%
Testing Accuracy	78.60%
Precision	0.79
Recall	0.79
F1-score	0.79
Training Time	936.92 s
Total Parameters	14,848,586

18.2 Comparison of CNN Architectures

Model	Parameters	Accuracy (%)	Training Time
LeNet-5	83,126	60.86	50.79 s
AlexNet	6,976,842	73.84	197.84 s
VGG16	14,848,586	78.60	936.92 s
ResNet50	24,114,826	78.56	363.00 s

19. Discussion Questions

Answer the following questions.

- (a) Why is AlexNet considered a breakthrough in deep learning?
It proved deep CNNs could outperform hand crafted feature methods and also introduced GPU training
- (b) Why does VGG16 use only 3×3 convolution filters?
Stacking multiple 3×3 filters achieves the same receptive field as a larger filter but with fewer parameters and more non-linearities i.e. more ReLU activations between layers, improving both efficiency and representational power.
- (c) Explain the advantages of the Inception module.
Extracts multi-scale features (1×1 , 3×3 , 5×5 in parallel) 1×1 convolutions reduce dimensionality before expensive operations, cutting parameters/compute, More computationally efficient than a single large uniform architecture and achieves strong accuracy without excessive depth alone
- (d) What is the purpose of residual learning?
Solves the vanishing gradient problem in very deep networks by adding skip connections, letting the network learn a residual $F(x) = H(x) - x$ instead of the full mapping directly.
- (e) Differentiate LeNet and ResNet.
LeNet-5 ResNet Depth 7 layers Up to 152+ layers Parameters 60K Millions Key idea Basic conv+pool stack Skip/residual connections Problem solved Feature learning vs handcrafted Vanishing gradients in deep nets Use case Simple digit recognition Large-scale image classification
LeNet is a simple digit recognition model with 7 layers with basic conv and pooling with around 60k parameters whereas Resnet is an image classification model with skip/residual connections with 152 layers at around millions of parameters
- (f) What is Transfer Learning?
Reusing a CNN pretrained on a large dataset (e.g., ImageNet) for a new, often smaller task — by keeping its learned convolutional features and replacing/retraining only the classifier layers, avoiding training from scratch.
- (g) Why is fine tuning required?
Pretrained features are general but may not always be optimal for the new dataset's specific classes. Fine-tuning adapts those features to be more task-specific, improving accuracy beyond just training a new classifier head alone.
- (h) Explain the difference between dilated convolution and transpose convolution.
Dilated convolution: expands the receptive field by inserting gaps between kernel elements, without increasing parameters or reducing spatial resolution — used to see more context (e.g., segmentation). Transpose convolution: performs learnable upsampling, increasing spatial dimensions of a feature map — used to reconstruct/generate larger outputs from smaller ones (e.g., GANs, autoencoders, segmentation output).
- (i) Why do pretrained models converge faster?
They start with already-learned, generalizable low/mid-level features instead of random weights, so the network doesn't need to relearn these basics.

- (j) Compare the computational complexity of LeNet and ResNet.
LeNet-5 is extremely lightweight (60K params, shallow, fast on CPU, minimal FLOPs) — suitable only for simple tasks like digit recognition. ResNet (e.g., ResNet50, 25M params) is far deeper and computationally heavier, requiring GPUs and significantly more FLOPs per inference, but achieves vastly higher accuracy on complex, large-scale datasets — a direct trade-off of compute cost for representational power.

20. SOURCE CODE

<https://github.com/sriwhatsnext/dllab/tree/main/Week-04-TransferLearning>

21. References

- (a) Yann LeCun, Leon Bottou, Yoshua Bengio and Patrick Haffner, *Gradient-Based Learning Applied to Document Recognition*, Proceedings of the IEEE, 1998.
- (b) Alex Krizhevsky, Ilya Sutskever and Geoffrey Hinton, *ImageNet Classification with Deep Convolutional Neural Networks*, NeurIPS, 2012.
- (c) Karen Simonyan and Andrew Zisserman, *Very Deep Convolutional Networks for Large-Scale Image Recognition*, ICLR, 2015.
- (d) Christian Szegedy et al., *Going Deeper with Convolutions*, CVPR, 2015.
- (e) Kaiming He, Xiangyu Zhang, Shaoqing Ren and Jian Sun, *Deep Residual Learning for Image Recognition*, CVPR, 2016.
- (f) Ian Goodfellow, Yoshua Bengio and Aaron Courville, *Deep Learning*, MIT Press, 2016.
- (g) TensorFlow Documentation: <https://www.tensorflow.org>
- (h) Keras Documentation: <https://keras.io>